LockTrace (BCT) Technical Whitepaper

A High-Throughput Hybrid Proof-of-Work, 3-Phase Atomic CAS Sequencer & Quorum-Attested Layer-1 Blockchain for Censorship-Resistant SocialFi, Decentralized Naming (.bct), and Self-Sovereign Cross-Chain DeFi

Protocol Release: v2.9.3 (Production Mainnet)	**Release Authority:** BCT885bc383a03ebed0aa11cedfbfac0d52be3d4080
Official Mainnet: https://www.locktrace.space	**Bridge Proof-of-Reserve Vault:** BCT_BRIDGE_VAULT_0000000000000000000000000000000000
Base Mainnet wBCT ERC-20: 0xA3C743113a41cFd0146917065d4578337Bb77254	**Validator Staking Vault:** BCT_STAKE_VAULT_0000000000000000000000000000000000

A High-Throughput Hybrid Proof-of-Work, 3-Phase Atomic CAS Sequencer & Quorum-Attested Layer-1 Blockchain for Censorship-Resistant SocialFi, Decentralized Naming (.bct), and Self-Sovereign Cross-Chain DeFi

LockTrace Core Architecture Team & Decentralized Validator Community

Official Mainnet: https://www.locktrace.space

Abstract

Modern digital communication, identity naming, and financial settlement remain heavily monopolized by centralized platforms that impose unilateral censorship, algorithmic rent-seeking, custodial insolvency risks, and severe latency bottlenecks. Conversely, legacy Layer-1 blockchains suffer from either prohibitive gas fees, sequential execution bottlenecks that freeze user interfaces during block production, or plutocratic validator cartels.

LockTrace (Ticker: BCT) introduces a mathematically provable, high-concurrency Layer-1 blockchain protocol engineered from the ground up to unify **censorship-resistant on-chain social networking (LockTrace Space)**, **sovereign decentralized identity (`.bct` Blockchain Naming System)**, and **non-custodial decentralized finance (DeFi)** within a single zero-trust state machine.

In **Version 2.9.3**, LockTrace establishes sovereign peer-to-peer resilience, ultra-efficient block propagation, decentralized hardware acceleration, and comprehensive multi-layer security hardening:

- **High Start-Nonce Rayon Clamping (AUD16-01):** Clamped `max_nonces` in `crates/locktrace_core/src/pow.rs` and bounded 64-bit integer ranges to eliminate empty chunk Rayon iterations and 100% CPU thread exhaustion.
- **Prioritized Non-Empty Database Discovery (AUD16-02):** Reordered candidate database paths in `scripts/test_base_logs.py` to prioritize active `/opt/blockchaint/data/blockchain.db` over stale 0-byte databases.
- **DOM XSS Attribute Sanitization (AUD16-03):** Replaced `escapeHtml` with `sanitizeJsAttr` across all inline `onclick` event attributes in `scripts/replace_embed.py`, `frontend/index.html`, and `templates/index.html`.
- **P2P Compact Block Header & Duplicate ID Defense (AUD16-04):** Enforced non-empty string validation on `CompactHeader` fields (`hash`, `previous_hash`, `merkle_root`, `miner`) and eliminated duplicate transaction IDs via `HashSet` in `crates/locktrace_core/src/compact.rs`.
- **Thirdweb 64-Hex Validation & Dual-Prefix Key Redaction (AUD16-05):** Strictly validated 64-hex private keys, ensured lowercase `0x` formatting in `thirdweb_engine.mjs`, and masked both raw 64-hex and `0x`/`0X` prefixed keys in Python log redaction.
- **BIP-39 Pre-Split NFKD Unicode Normalization (AUD16-06):** Applied NFKD normalization directly to raw mnemonic strings in `core/mnemonic.py` prior to whitespace splitting, fully conforming to BIP-39 specification.
- **Audit Bundle Privacy Filters & Output XML Parity (AUD16-07):** Filtered `"wallet"` and `".bak"` artifacts from security packages and ensured dynamic `repomix-output_16.xml` bundling.
- **Universal Protocol SSOT Version Synchronization (AUD16-08):** Harmonized all deployment scripts, launchers, whitepapers, and core packages to v2.8.9.

Nginx Child Location CSP Inheritance & /integrity.json Parity (AUD15-04): Re-declared Content-Security-Policy across all sub-locations with custom `add_header` and added explicit `/integrity.json` caching guards in `nginx.conf`.

Full BIP-39 NFKD Normalization Prior to Validation (AUD15-05): Enforced Unicode NFKD normalization in both `validate_mnemonic` and `mnemonic_to_private_key` before wordlist verification.

P2P Compact Block Version, Raw JSON & Short ID Verification (AUD15-06): Enforced `compact_version == 1`, 64KB raw JSON bounds, and 16-hex character formatting in `crates/locktrace_core/src/compact.rs`.

Thirdweb 0x Key Normalization & CLI Flag Injection Defense (AUD15-07): Normalized raw 64-hex server private keys with `0x` prefix and prohibited leading hyphens in transaction ID and chain name regexes.

Audit Bundle Privacy Filters & Dynamic Protocol SSOT (AUD15-08): Configured strict exclusions for `.env`, keys, certificates, and bound protocol version dynamically to `Config.PROTOCOL_VERSION`.

Live Coinbase Spot Price Feed CSP & Nginx Header Preservation (NEW-02 & NEW-03): Whitelists `https://api.coinbase.com` in `connect-src` and preserves HSTS/nosniff/X-Frame-Options across static PWA and Service Worker location blocks.

Anti-Spoofing Cryptographic Quote Attestation (NEW-04): Removes misleading 'verified' account badges from plain text quote blocks, restricting verified badges strictly to cryptographically proven on-chain identities.

WrappedBCT Un-Indexed Event Log Parity (NEW-05): Updates `MintedFromBridge` to emit un-indexed L1 transaction hashes, ensuring raw hash strings are preserved in EVM logs for cross-chain explorers and indexers.

Atomic Reference-Swap Multithreaded State Indexes (NEW-06): Replaces in-place dictionary clears with atomic reference swaps in `_load_in_memory_bns_index` and `_load_in_memory_space_index`, guaranteeing race-free concurrent read access under multi-threaded Gunicorn serving.

Comment Reply Hyphenation & Robust .env Parser (NEW-07): Supports hyphenated BNS handles in reply cancellation and cleanly strips inline comments during `.env` loading.

Zero-Bypass Public Key Address Derivation: Prohibits unauthenticated double-SHA256 address generation; invalid or off-curve public keys return `None` immediately across Rust native engine and Python fallback.

Strict Exception-Based Merkle Duplicate Rejection (CVE-2012-2459): Raises explicit `ValueError` on duplicate transaction leaves, preventing block tampering and eliminating `"0"*64` Merkle root collision exploits.

Single-Transaction Block SPV Proof Parity: Fully supports zero-step SPV inclusion proofs for single-transaction blocks where the transaction hash directly equals the block Merkle root.

Cryptographic Sentinel Nonce Serialization (LT-POW-01): Replaces static split tokens with 128-bit random runtime sentinels (`secrets.token_hex(16)`) and exact byte-offset splicing across `core/gpu_miner.py` and `core/native_bridge.py`, immunizing the PoW mining engine against transaction-content split DoS exploits.

Strict SPV Merkle Proof Validation (LT-MRK-01): Enforces 64-character lowercase hex format validation on target hashes and Merkle roots, rejects empty proofs with mismatched roots, and strictly verifies directional paths (`"left"` vs `"right"`) in both Python and Rust (`crates/locktrace_core/src/merkle.rs`).

SECP256k1 Curve Point Validation & 33-Byte Compressed Parity (LT-CRY-01): Adds strict SEC1 curve verification in Rust (`k256::AffinePoint::from_encoded_point`) and Python fallback, rejecting off-curve points ($y^2 \neq x^3 + 7$) and providing full address derivation parity for 33-byte compressed public keys.

Cross-Chain Canonical Address Ingress Normalization (LT-SOL-02): Enforces case-insensitive canonical address mapping (`BCT` + lowercase hex) across `core/bridge.py`, `server.py`, and `core/bridge_relayer.py`, preventing unclaimable bridged assets.

Multi-Core Parallel Thread Mining & Honest Telemetry (LT-GPU-01): Integrates `ThreadPoolExecutor` parallel worker pool and first-priority delegation to Rust Rayon in `core/gpu_miner.py` with accurate engine telemetry.

Native Rust Acceleration Core (`crates/locktrace_core`): High-speed cryptographic validation, lazy parallel Rayon PoW mining with accurate atomic hashrate telemetry, zero heap allocation chunking, and deterministic serialization.

Multi-Layer Cross-Chain Bridge Hardening (LT-SOL-01): Hex-character enforcement (`[0-9a-fA-F]`) on 43-character BCT addresses and zero burn address rejection across Solidity smart contract (`WrappedBCT.sol`), frontend Web3 client, and backend bridge relayer daemon.

ECDSA Space Cast Authentication (LT-WEB-01): Enforces cryptographic secp256k1 canonical signature verification on quick compose casts, eliminating unauthorized impersonation.

DOM XSS Sanitization (LT-WEB-02): Strict regex attribute sanitation and entity escaping on mentions, hashtags, and cashtags in rich social threads.

Nginx HTTPS Enforcement & Tight CSP (LT-NGX-01): Unconditional port 80 HTTP-to-HTTPS 301 redirection and tightened Content-Security-Policy removing `unsafe-eval` and whitelisting trusted RPC origins.

Thirdweb Engine EVM Address & BigInt Validation (LT-TWB-01): Defensive validation against negative numbers, non-integers, and invalid EVM address formats.

P2P Compact Block Relay (v2.6.0): Transmits block headers, prefilled coinbase/system transactions, and 16-hex short transaction IDs, cutting network bandwidth consumption by ~90-95% while strictly enforcing cryptographic Merkle root and PoW verification.

Decentralized State Snapshot Sync (v2.7.0): Enables warp sync for newly joining nodes by providing deterministically hashed state roots over sorted account balances and BNS registries, requiring nodes to fully replay only the rolling window of recent 1,000 blocks to tip without compromising decentralization.

Hybrid OpenCL GPU & Multi-Worker CPU Miner Engine (v2.7.0): Integrated OpenCL C acceleration for NVIDIA, AMD, and Intel GPUs alongside multi-worker CPU fallback in `node_runner.py` with zero required external pip packages.

Emergency LWMA Stall Failsafe (v2.5.0): Dynamic difficulty relaxation when block intervals exceed 120s (6x target block time), preventing network deadlocks during severe miner drop-offs.

Priority Replace-By-Fee (RBF) & Mempool Eviction (v2.5.0): Dynamic mempool admission that automatically evicts the lowest-fee transaction in saturated mempools (5,000 txs) for higher-fee priority inclusion, rendering mempool flood DoS economically unviable.

Constant-Time SECP256k1 Cryptography Engine (v2.5.0): Full migration to OpenSSL C-backed constant-time cryptography eliminating Minerva timing side-channels (CVE-2024-23342).

O(1) Reverse-Cursor Transaction Indexing & Query Scalability (v2.5.0): Zero-overhead paginated transaction lookup and indexed SQLite access eliminating linear memory/CPU degradation.

Consensus Quorum Proof-of-Attestation & Fake Uptime Immunity (LT-01): Eliminates Sybil stake spoofing and idle heartbeat emission exploitation by strictly requiring both collateral locked in `BCT_STAKE_VAULT_...` (minimum 1,000.0 BCT) and real-time cryptographic ECDSA block attestations within the last 3 blocks for 30% validator pool reward participation.

Cross-Chain Bridge Reorg & Concurrency Guard (LT-02): Enforces 8-to-10 block L1 confirmation depth prior to Base EVM minting, strict *fail-closed* 12-block Base confirmation depth verification, SQLite unique index constraints on EVM burn hashes, and in-flight mutex locks preventing double-spend race conditions.

BIP-62/146 Low-S Signature Malleability Immunity & Deterministic Canonical Signatures (LT-04): Strictly enforces canonical ECDSA Low-S ($s \leq \lfloor N/2 \rfloor$) in SECP256k1 signature validation, rejecting all malleable signatures, and mandates deterministic canonical payload templates (`TX:mainnet:<sender>:<receiver>:<amount>:<nonce>`) with strictly sequential nonces.

Zero-Disk TOCTOU-Free Atomic In-Memory Serving (LT-07): Serves `index.html`, `sw.js`, and `manifest.json` directly from cryptographically verified in-memory byte buffers, eliminating Time-of-Check to Time-of-Use tampering risks.

Table of Contents

[Introduction & Architectural Motivation](#1-introduction--architectural-motivation)

[v2.7.0 High-Concurrency Engine: 3-Phase Atomic CAS & O(1) Delta WAL](#2-v270-high-concurrency-engine-3-phase-atomic-cas--o1-delta-wal)

[Cryptographic Primitives & 10^8 Atomic Unit Accounting](#3-cryptographic-primitives--108-atomic-unit-accounting)

[Dual Hybrid Consensus: 70% PoW Proposer + 30% Validator Quorum](#4-dual-hybrid-consensus-70-pow-proposer--30-validator-quorum)

[Hybrid Time-and-Quota Batch Sequencer](#5-hybrid-time-and-quota-batch-sequencer)

[LockTrace Space: Native On-Chain SocialFi & Creator Royalties](#6-locktrace-space-native-on-chain-socialfi--creator-royalties)

[BCT Naming System (BNS): Sovereign `.bct` Identity & Marketplace](#7-bct-naming-system-bns-sovereign-bct-identity--marketplace)

[Tokenomics, Deflationary Mechanics & Supply Invariants](#8-tokenomics-deflationary-mechanics--supply-invariants)

[Cross-Chain Interoperability: 1:1 Base Mainnet wBCT Bridge](#9-cross-chain-interoperability-11-base-mainnet-wbct-bridge)

[P2P Compact Block Relay (BIP-152 Architecture)](#10-p2p-compact-block-relay-bip-152-architecture)

[Decentralized State Snapshot Warp Sync Architecture](#11-decentralized-state-snapshot-warp-sync-architecture)

[Hybrid OpenCL GPU & Multi-Worker CPU Miner Engine](#12-hybrid-opencl-gpu--multi-worker-cpu-miner-engine)

[On-Chain Frontend Integrity Shield & Tamper Detection](#13-on-chain-frontend-integrity-shield--tamper-detection)

[Multi-Layered Cyber Defense: Anti-DDoS, Replay & Sybil `/24` Hardening](#14-multi-layered-cyber-defense-anti-ddos-replay--sybil-24-hardening)

[Decentralized Node Runner (`node_runner.py`) & P2P Topology](#15-decentralized-node-runner-node_runnerpy--p2p-topology)

[Comprehensive Threat Model & Formal Verification Matrix](#16-comprehensive-threat-model--formal-verification-matrix)

[Roadmap & Protocol Evolution](#17-roadmap--protocol-evolution)

1. Introduction & Architectural Motivation

1.1 Structural Failures of Legacy Web2 and Web3 Systems

Contemporary digital networks fail users across four critical dimensions:

- **Unilateral Platform Censorship:** Centralized social networks arbitrarily de-platform creators, shadow-ban discourse, and alter social graphs without cryptographic accountability.
- **Predatory Intermediary Economics:** Creators generate the entirety of user engagement on Web2 social networks yet receive zero direct protocol royalties from likes, comments, or reposts.
- **L1 Execution Stalls & UI Freezing:** Traditional single-threaded blockchain nodes lock ledger state during block assembly and persistence, causing severe request queueing, HTTP timeouts, and UI freezes when thousands of users transact simultaneously.
- **Unverified Client Supply-Chain Attacks:** Most Web3 decentralized applications (DApps) serve unverified JavaScript from centralized CDNs; a single DNS hijack or compromised CDN cache allows attackers to drain non-custodial wallets silently.

1.2 The LockTrace v2.7.0 Solution

LockTrace v2.7.0 resolves these challenges through an end-to-end verifiable stack:

- **Protocol-Level SocialFi & Naming:** Posts, quote-reposts, likes, comments, follows, and `.bct` domain operations are first-class Layer-1 transactions—eliminating smart-contract reentrancy bugs while executing instant on-chain royalty splits.
- **Lock-Free Asynchronous PoW & O(1) Delta Storage:** High-frequency user actions never wait on PoW mining or full-chain disk writes, enabling thousands of concurrent users to interact with sub-5ms responsiveness.
- **P2P Compact Block Relay & Warp Snapshot Sync:** Blocks propagate with 90-95% less bandwidth, while new nodes warp sync in seconds using deterministic state roots and a 1,000-block rolling verification window.
- **Zero-Knowledge Browser Custody + On-Chain Code Verification:** Private keys never leave the client's WebCrypto sandbox, and the client verifies its own SHA-256 + ECDSA release signature against the blockchain before unlocking wallet operations.

1.3 Zero-Custodial Security Policy (v2.7.0 Mandate)

A core security breakthrough in Version 2.7.0 is the absolute elimination of custodial and hybrid key-handling vectors:

- **Zero Transmission of Secret Material:** Direct transmission of mnemonic phrases or private keys over HTTP/REST is strictly forbidden and rejected with `HTTP 400 Zero-Custodial Security Violation`.
- **Client-Side SECP256k1 ECDSA Execution:** All state-changing operations (transfers, staking, unstaking, Space posts, likes, comments, follows, BNS registrations) are signed exclusively inside the client's browser or local terminal memory using RFC 6979 deterministic SECP256k1 signatures.
- **Decentralized Relay Isolation:** Bridge relayer and node authority credentials are held strictly within isolated daemon environments with encrypted keystores; no shared or static key files remain in public repositories.

2. v2.7.0 High-Concurrency Engine: 3-Phase Atomic CAS & O(1) Delta WAL

A foundational breakthrough in LockTrace v2.7.0 is the elimination of `O(N)` full-chain serialization and synchronous lock contention under heavy multi-user load.

```
+-------------------------------------------------------------------------------+
|              LOCKTRACE v2.7.0 NON-BLOCKING TRANSACTION & CAS PIPELINE          |
+-------------------------------------------------------------------------------+
```

2.1 Why Legacy Blockchain Nodes Freeze Under Load

In naive implementations, reaching a batch threshold inside a user's HTTP request triggers synchronous Proof-of-Work mining while holding the global ledger mutex, followed by serializing the entire blockchain (`[b.to_dict() for b in chain]`) to disk. At 10,000+ blocks and 1,000+ concurrent users, this creates **O(N)** disk I/O amplification and multi-second request stalls.

2.2 3-Phase Atomic Compare-And-Swap (CAS) Block Sealing

LockTrace v2.7.0 separates transaction admission from block sealing via a strict 3-phase protocol:

Phase 1 — Canonical Snapshot: Under `self.lock`, the sequencer captures:

$$\text{Snapshot} = \left(h_{expected} = |\mathcal{C}|,\; H_{prev} = \mathcal{C}[-1].hash,\; \mathcal{P} = \{tx.hash() \mid tx \in Mempool\}\right)$$

along with deterministic 70/30 reward transactions.

Phase 2 — Lock-Free Proof-of-Work: Guarded by `self._mining_lock` (preventing redundant local PoW threads) but **outside** `self.lock`, the worker searches for a nonce satisfying *SHA256(Header) < Target(D)* using OpenCL GPU vectorization or multi-threaded CPU. During this entire window, HTTP workers process reads and new transactions with zero lock wait.

Phase 3 — Atomic CAS Commit & Fork Guard: Upon solving the puzzle, the worker re-acquires `self.lock`, synchronizes with SQLite WAL, and evaluates the Compare-And-Swap invariant:

$$\text{CommitAllowed} \iff \left(|\mathcal{C}| == h_{expected}\right) \land \left(\mathcal{C}[-1].hash == H_{prev}\right) \land \left(\mathcal{P} \subseteq Mempool_{live}\right)$$

- **Zero Chain Split (Fork Immunity):** If an external PoW miner (`submit_mined_block`) or P2P peer advanced the chain while Phase 2 was hashing, the CAS condition evaluates to `False` and the local candidate is deterministically discarded (`[CAS FORK GUARD]`).
- **Zero Transaction Loss:** Upon commit, the mempool is pruned strictly via set difference:

$$Mempool_{next} = \{tx \in Mempool_{live} \mid tx.hash() \notin \mathcal{P}\}$$

preserving 100% of transactions that arrived while PoW was in progress.

2.3 O(1) Delta SQLite WAL Persistence & Multi-Worker Sync

- **Delta Block Writes:** `SQLiteStorage.get_max_block_index()` queries the highest persisted `block_index` in `O(1)` time. Only blocks with `index > max_db_idx` are serialized and inserted into the `blocks` and `tx_lookup` tables.
- **Dirty Wallet Upserts:** The engine tracks `_saved_wallet_nonces` as `(public_key, nonce)` tuples per address, writing only mutated accounts (`dirty_wallets`) per transaction rather than rewriting the entire wallet registry.

- **O(\Delta) Cross-Worker Synchronization (`load_delta_state`):** When Gunicorn worker processes detect a bumped `state_revision`, they verify their local tip hash against `SELECT hash FROM blocks WHERE block_index = ?` and fetch only `WHERE block_index > ?`, reducing multi-process synchronization overhead from **O(|\mathcal{C}|)** to **O(\Delta)**.

2.4 Emergency LWMA Stall Failsafe (v2.5.0)

If the network encounters an acute miner drop-off causing the elapsed time between the tip and current time to exceed *6 × T_{target} = 120 seconds*, the Linear Weighted Moving Average algorithm dynamically relaxes target difficulty:

If (t_{current} - t_{tip}) > 120s \implies D_{effective} = \max≤ft(1, \lfloor D_{LWMA} / 2 \rfloor\right)

This decentralized failsafe guarantees that the blockchain cannot permanently freeze during severe hashrate declines without requiring hard forks or human intervention.

2.5 Priority Replace-By-Fee (RBF) & Mempool Eviction (v2.5.0)

When the mempool reaches its maximum capacity ceiling (`MAX_GLOBAL_MEMPOOL_SIZE = 5,000`), incoming transactions with a higher fee density evict the lowest-fee unconfirmed transaction:

Evict tx_{\min} \iff fee(tx_{in}) > fee(tx_{\min}) \land |Mempool| ≥ 5000

This renders spam-flooding attacks economically unviable, as an attacker attempting to monopolize mempool space must continually outbid authentic economic transactions.

---

3. Cryptographic Primitives & *10^8* Atomic Unit Accounting

3.1 SECP256k1 Elliptic Curve Digital Signatures & Constant-Time OpenSSL Engine

All account identities, transactions, node heartbeats, block submissions, and frontend releases are authenticated via the Koblitz curve `secp256k1`:

y^2 \equiv x^3 + 7 \pmod p, \quad p = 2^{256} - 2^{32} - 977

- **Constant-Time Verification:** All ECDSA operations leverage C-backed OpenSSL primitives (`cryptography` library), ensuring constant-time signature verification that strictly eliminates Minerva side-channel timing attacks (CVE-2024-23342).
- **Deterministic Signing (RFC 6979):** Signatures use HMAC-SHA256 deterministic *k*-nonce generation, eliminating random number generator (CSPRNG) entropy failures.
- **Address Derivation:** Given an uncompressed 64-byte public key *Q = X \parallel Y*:

Address = "BCT" \parallel SHA256≤ft(SHA256(Q)_{hex}\right)[0..39]

yielding a 43-character collision-resistant address (e.g., `BCT885bc383a03ebed0aa11cedfbfac0d52be3d4080`).

3.2 BIP-62/146 Low-S Signature Malleability Immunity

Every valid ECDSA signature consists of a pair *(r, s)*. Because *(r, s)* and *(r, -s \pmod N)* are both mathematically valid for curve order *N*, malicious relays can mutate transaction hashes without invalidating signatures. LockTrace v2.7.0 strictly rejects non-canonical signatures:

Accept(s) \iff s ≤ \lfloor N/2 \rfloor

All client libraries and WebCrypto signers enforce Low-S prior to transmission.

3.3 Binary SHA-256 Merkle Tree & SPV Inclusion Proofs

Transactions within each block are committed into a cryptographic binary Merkle tree. Light clients verify transaction inclusion via Simplified Payment Verification (SPV) using **O(\log_2 N)** audit paths:

MerkleRoot = TreeHash≤ft(H(tx_0), H(tx_1), \dots, H(tx_{n-1})\right)

where odd leaves are duplicated at each layer.

3.4 *10^8* Atomic Unit Accounting (Zero Drift Invariant)

To prevent floating-point rounding errors and precision degradation at a 1-Trillion scale, all internal state calculations operate in integer atomic units:

1 BCT = 10^8 atomic units = 100,000,000 units

Floating-point arithmetic is strictly prohibited in consensus validation, balance assertions, and reward accounting.

---

4. Dual Hybrid Consensus: 70% PoW Proposer + 30% Validator Quorum

LockTrace implements a hybrid consensus model that balances the decentralized permissionlessness of Proof-of-Work with the high-speed finality checkpoints of an active validator quorum.

4.1 Block Reward Allocation Formula

For each successfully mined and verified block B_k, the gross reward pool consists of the block mining subsidy plus distributed transaction fees:

$$R_{total}(B_k) = R_{subsidy} + \sum_{tx \in B_k} Fee_{reward}(tx)$$

This reward pool is split deterministically:

- **70% Block Proposer (PoW Miner):** $R_{proposer} = \lfloor 0.70 \times R_{total}(B_k) \rfloor$
- **30% Active Quorum Validators:** $R_{quorum} = \lfloor 0.30 \times R_{total}(B_k) \rfloor$

4.2 Proof-of-Attestation Participation Invariant (LT-01)

To qualify for the 30% Quorum Pool, a node must satisfy two cryptographic conditions:

Locked Staking Vault Collateral: The node's address must have at least **1,000.0 BCT** locked in `BCT_STAKE_VAULT_...`.

Recent Cryptographic Attestation: The node must have submitted an ECDSA-signed attestation (`/api/v1/nodes/attest`) for a block within a 3-block rolling window ($last_block_height \geq tip - 3$).

Nodes failing either criterion are pruned from quorum payout calculation.

4.3 Staking Vault & Non-Custodial Slashing Mechanism

Validators lock collateral via a 100% non-custodial transaction to `BCT_STAKE_VAULT_...`. Unstaking requires an explicit withdrawal request with a 24-hour unbonding cooldown.

- **Equivocation Slashing:** If a validator signs two conflicting block proposals or attestations at the identical block height, any node can submit cryptographic proof of double-signing, triggering an on-chain **50% stake slash** (*500.0 BCT* permanently burned to `BCT0000000000000000000000000000000000000000`) and immediate committee eviction.

---

5. Hybrid Time-and-Quota Batch Sequencer

LockTrace optimizes network throughput and mempool efficiency through categorized, deterministic micro-batching windows:

- **Social Likes (`space_like`):** Window $\tau = 20s$ or Quota $Q = 15$ txs.
- **Social Comments (`space_comment`):** Window $\tau = 50s$ or Quota $Q = 10$ txs.
- **General Transfers & Naming:** Window $\tau = 20s$ or Quota $Q = 25$ txs.

When either elapsed time $\Delta t \geq \tau$ or pending count $N \geq Q$ is reached, the sequencer triggers `_sequencer_wakeup`, packaging the batch into an asynchronous CAS block candidate without stalling client HTTP threads.

---

6. LockTrace Space: Native On-Chain SocialFi & Creator Royalties

LockTrace Space embeds social interactions directly into the Layer-1 state machine, guaranteeing censorship resistance and creator royalties.

6.1 Action Fee & Distribution Schedule

Publish Post (`space_post`):

• Protocol Fee: *0.10 BCT*

• Distribution: *80\%* (*0.08 BCT*) returns to Treasury Vault; *20\%* (*0.02 BCT*) distributed to miners and validators.

• Canonical Signature Template: `{author_address}:{content}`

Like Creator (`space_like`):

• Cost: *0.01 BCT*

• Distribution: *+0.002 BCT* (*20\%*) paid directly to the post author's wallet; *0.007 BCT* (*70\%*) returns to Treasury; *0.001 BCT* (*10\%*) rewards validators.

• Canonical Signature Template: `{liker_address}:{post_id}`

Comment Royalty (`space_comment`):

• Cost: *0.02 BCT*

• Distribution: *+0.004 BCT* (*20\%*) paid directly to the original author; *0.014 BCT* (*70\%*) returns to Treasury; *0.002 BCT* (*10\%*) rewards validators.

• Canonical Signature Template: `{commenter_address}:{post_id}:{content}`

Follow Creator (`space_follow`):

• Cost: *1.00 BCT*

• Distribution: *+0.50 BCT* (*50\%*) deposited directly to the target creator's wallet; *0.40 BCT* (*40\%*) locks into Treasury; *0.10 BCT* (*10\%*) rewards validators.

• Canonical Signature Template: `{follower_address}:{target_address}`

---

7. BCT Naming System (BNS): Sovereign `.bct` Identity & Marketplace

The BCT Naming System provides decentralized, human-readable identity handles (`@name.bct`) mapped to cryptographic addresses.

7.1 Tiered Character Pricing

To prevent domain hoarding, registration costs scale inversely with character length:

• **3-Letter Domains (e.g., `btc.bct`):** *100.00 BCT*

• **4-Letter Domains (e.g., `sats.bct`):** *50.00 BCT*

• **5+ Letter Domains (e.g., `locktrace.bct`):** *10.00 BCT*

Registration fees follow an **80% Treasury Return / 20% Miner Distribution** split. Domains include an integrated on-chain secondary marketplace supporting non-custodial buy, sell, and price listing operations.

---

8. Tokenomics, Deflationary Mechanics & Supply Invariants

8.1 Total Capped Supply

The maximum lifetime supply is hard-capped at **1,000,000,000,000.00000000 BCT** (1 Trillion BCT). New token minting is mathematically and architecturally prohibited.

8.2 Deflationary Treasury Return (Havuza Dönü■)

Unlike inflationary networks that dilute holders, LockTrace channels 50% of all standard transfer fees, 80% of Space post and BNS registration fees, and 70% of social interaction fees back into the Genesis Treasury Vault (`BCT_HAZINE_...`). This reduces circulating velocity and strengthens token scarcity with every network interaction.

8.3 Welcome Faucet Anti-Sybil Bounding

New wallet creation bonus (*1.00 BCT*) is disbursed directly from the Treasury Vault under strict cryptographic bounds:

• Per-IP rate limiting: maximum 3 creations per minute, 10 per hour.

- Global rolling ceiling: maximum 25 grants per hour network-wide.

9. Cross-Chain Interoperability: 1:1 Base Mainnet wBCT Bridge

LockTrace connects to Ethereum Layer-2 via a decentralized 1:1 asset bridge to the Base Mainnet Wrapped BCT contract (`0xA3C743113a41cFd0146917065d4578337Bb77254`).

9.1 Bridge Proof-of-Reserve Invariant

Supply_{wBCT}(Base) \equiv Balance_{native}(BCT_BRIDGE_VAULT_...)

Every minted wBCT is backed 1:1 by native BCT escrowed on Layer-1. Arbitrary minting without matching L1 deposit is cryptographically blocked.

9.2 Reorg Depth & Mutex Concurrency Guards (LT-02)

- **L1 to Base Minting:** Relayers require a minimum confirmation depth of **8-to-10 blocks** on LockTrace Layer-1 prior to initiating an EVM mint transaction.
- **Base to L1 Unlocking:** Relayers require **12 confirmed Base blocks** before releasing native BCT from `BCT_BRIDGE_VAULT_...`.
- In-flight mutex locks and SQLite unique transaction indices prevent race conditions and double-spending across bridge operations.

10. P2P Compact Block Relay (BIP-152 Architecture)

In Version 2.6.0, LockTrace implements BIP-152 style **Compact Block Relay** to eliminate bandwidth bottlenecks during block propagation.

10.1 Short Transaction ID Hashing Scheme

Rather than transmitting full transaction bodies across the P2P gossip network, nodes compute 16-hex short transaction IDs:

K_{short} = SHA256(HeaderBytes \parallel nonce)[0..15]

A compact block message transmits:

Canonical Block Header (112 bytes).

Prefilled Transactions (Coinbase, 70/30 reward distributions, system transactions).

Vector of 16-character Short Transaction IDs for all user transactions in the Merkle tree.

10.2 Reconstruction & Roundtrip Efficiency

Upon receiving a compact block:

The receiving node maps its local mempool transactions to their corresponding short IDs using the block's salt key *K_{short}*.

If all transactions are present in the local mempool, the node reconstructs the full block in < 1 *ms* with zero network roundtrips.

If transactions are missing, the node sends a lightweight `/p2p/getblocktxn` request specifying the missing indices.

Bandwidth Savings: A standard 1,000-transaction block (~250 KB) is reduced to a compact payload of ~7.5 KB—achieving a **>95% bandwidth reduction**.

11. Decentralized State Snapshot Warp Sync Architecture

In Version 2.7.0, LockTrace introduces **Decentralized State Snapshot Warp Sync**, allowing new nodes to join and reach the active tip in seconds rather than hours.

11.1 Deterministic State Root Calculation

The blockchain state root is computed deterministically over lexicographically sorted accounts and active BNS records:

$$\mathcal{A} = Sorted\left(\{ (addr_i, bal_i, nonce_i) \mid bal_i > 0 \lor nonce_i > 0 \}\right)$$

$$\mathcal{B} = Sorted\left(\{ (dom_j, owner_j, expires_j) \}\right)$$

$$R_{state} = SHA256\Big(SHA256(\mathcal{A}) \parallel SHA256(\mathcal{B})\Big)$$

11.2 1,000-Block Rolling Verification Window

To preserve full decentralization and trustlessness:

The joining node downloads the latest verified snapshot (`/api/v1/snapshot/latest`).

The node validates the deterministic state root against the attested block header at height $H_{snapshot} = tip - 1,000$.

The node downloads and fully replays the full transactions and PoW consensus checks of the **most recent 1,000 blocks** up to the live tip.

This eliminates historical chain replay overhead while guaranteeing 100% cryptographic consensus validation.

---

12. Hybrid OpenCL GPU & Multi-Worker CPU Miner Engine

Version 2.7.0 integrates native hardware acceleration for Proof-of-Work mining in `node_runner.py` with zero external pip dependencies:

- **OpenCL C Vectorized GPU Kernel (`core/gpu_miner.py`):** Offloads SHA-256 double-hashing and nonce increments across thousands of GPU compute cores on NVIDIA, AMD, and Intel graphics processors.
- **Multi-Worker CPU Fallback:** If no OpenCL GPU platform is available, `node_runner.py` automatically spawns multi-worker CPU mining threads using Python's standard `multiprocessing` library.
- **Dual-Engine Native Bridge (`crates/locktrace_core`):** Rust native acceleration modules provide high-throughput SIMD hashing with seamless fallback to C-backed OpenSSL.

---

13. On-Chain Frontend Integrity Shield & Tamper Detection

To protect non-custodial users from CDN tampering, DNS poisoning, and malicious proxy injection:

Every release of `frontend/index.html` and `sw.js` is hashed via canonical SHA-256 and IPFS CIDv1.

The release digest is signed by the Release Authority key (`BCT885bc...`) and sealed permanently on-chain in a `frontend_release` transaction.

Node servers inspect local frontend hashes before serving; any modification triggers an immediate `HTTP 403 Security Lock`.

Client browsers verify integrity on load; if any discrepancy is detected, a persistent **Security Alert Badge** is activated, warning the user immediately.

---

14. Multi-Layered Cyber Defense: Anti-DDoS, Replay & Sybil `/24` Hardening

LockTrace v2.7.0 incorporates multiple interlocking cyber defense shields:

Per-Wallet Mempool Quota (`MAX_PENDING_TX_PER_WALLET = 25`): Restricts single addresses from spamming the mempool.

Priority RBF Mempool Eviction: Protects against mempool saturation by evicting lowest-fee transactions on the 5,000-tx ceiling.

Emergency LWMA Failsafe: Halves difficulty after 120s stalls to prevent network deadlocks.

Bounded Sliding-Window Rate Limiter: Bounds memory tables (25,000 IP buckets, 50,000 sig cache entries) to prevent RAM exhaustion.

Full-History Signature Replay Immunity: O(1) set lookup over all committed and pending signatures guarantees zero transaction replay.

IPv4 `/24` & IPv6 `/48` Subnet Deduplication: Normalizes validator IPs to prevent cloud VM Sybil attacks from draining the 30% Quorum Pool.

SSRF-Hardened P2P Gossip: Rejects loopback, link-local, multicast, and RFC-1918 private IP endpoints.

15. Decentralized Node Runner (`node_runner.py`) & P2P Topology

`node_runner.py` is a single-file, zero-external-dependency Python 3 application serving as a full validator, miner, and local web wallet:

- **Local Sovereign Mode:** `python node_runner.py --local --open`
- **Warp Fast Sync Mode:** `python node_runner.py --local --sync-mode snapshot --open`
- **GPU Mining Node:** `python node_runner.py --server --port 5050 --gpu --wallet BCT...`
- **Public Validator RPC:** `python node_runner.py --server --port 5050 --wallet BCT...`

16. Comprehensive Threat Model & Formal Verification Matrix

Attack / Failure Vector	Threat Mechanism	LockTrace v2.7.0 Defense Mechanism	Complexity / Bound
Mining Stall / Chain Deadlock	Sudden drop-off in active mining hashrate	**Emergency LWMA Failsafe** dynamically halves target difficulty if interval > 120s	**O(1) Recovery**
Mempool Saturation Spam	Botnet flooding 10,000+ zero-fee transactions	**Priority Replace-By-Fee (RBF)** evicts lowest-fee txs on 5,000 ceiling + 25-tx per-wallet cap	**O(\log N) Eviction**
P2P Bandwidth Spike / Congestion	High transaction volumes bloating block gossip	**P2P Compact Block Relay (BIP-152)** transmits 16-hex short IDs, cutting payload > 90%	**O(1) Propagation**
Initial Node Sync Latency	Downloading full historical chain from Genesis	**State Snapshot Warp Sync** restores state root, replaying only recent 1,000 blocks to tip	**Seconds vs Hours**
Concurrent PoW & P2P Race (Fork)	Local sequencer mines while peer submits same height	**3-Phase Atomic CAS Fork Guard** verifies tip height and previous hash before commit	**0% Fork Probability**
Minerva Timing Attack	Side-channel timing variance in ECDSA signature verification	**Constant-Time OpenSSL SECP256k1** C-backed verification (CVE-2024-23342 mitigation)	**Constant Time**
Datacenter Sybil Quorum Attack	Spawning 100 cloud nodes in same subnet to drain 30% pool	`_extract_node_host_key` normalizes to IPv4 `/24` & IPv6 `/48` + wallet deduplication	**Max 1 Vote / Subnet**
Signature Malleability	Relay inverting signature *(r, -s \pmod N)*	**BIP-62/146 Low-S Enforcement** (*s* ≤ *\lfloor N/2 \rfloor*) strictly enforced on all transactions	**Deterministic *s***
Frontend CDN Hijacking	Compromised web server delivering malicious JS	**On-Chain Release Sealing** + Node SHA-256 lockout + Browser Persistent Security Alert	**HTTP 403 + UI Alert**

17. Roadmap & Protocol Evolution

- **v1.0 – v1.6 (Completed):** Mainnet Genesis, 70/30 PoW + Validator Quorum consensus, LockTrace Space SocialFi royalties, `.bct` BNS registry & marketplace, Base Mainnet `wBCT` 1:1 cross-chain bridge, Uniswap v4 liquidity pool.

- **v1.9 (Completed):** On-chain SHA-256 + IPFS CIDv1 Frontend Release Integrity Shield, Quote-Repost attribution cards, Explorer BNS/wallet full-history indexing, Persistent Mainnet Security Discrepancy Alert System.
- **v2.2.0 (Completed):** 3-Phase Atomic CAS Non-Blocking Block Producer, **O(1)** Incremental Delta SQLite WAL Engine, **O(\Delta)** Multi-Worker Sync, Anti-DDoS Per-Wallet Mempool Quotas, IPv4 `/24` Sybil Subnet Hardening, and Stat-Memoized Integrity Gate.
- **v2.5.0 (Completed):** Anti-Stall Emergency LWMA Failsafe (120s difficulty relaxation), Priority Replace-By-Fee (RBF) Mempool Admission & Eviction, Constant-Time SECP256k1 Cryptography Engine (OpenSSL C-backed, Minerva mitigation), **O(1)** Reverse-Cursor Transaction Indexing & Query Scalability.
- **v2.6.0 (Completed):** P2P Compact Block Relay (BIP-152 standard, short 16-hex txids, ~90-95% network bandwidth reduction, roundtrip reconstruction via `/p2p/getblocktxn`).
- **v2.7.0 (Completed):** Decentralized State Snapshot Warp Sync (deterministic state root over sorted accounts & BNS records, rolling 1,000-block verification), Hybrid OpenCL GPU & Multi-Worker CPU Miner Engine (`core/gpu_miner.py`, `node_runner.py`).
- **v2.8.0 (Completed):** Native Rust Acceleration Core (`crates/locktrace_core`), Multi-Layer Bridge Hardening (LT-SOL-01), ECDSA Space Cast Authentication (LT-WEB-01), DOM XSS Attribute Sanitization (LT-WEB-02), Nginx Strict HTTPS Enforcement & CSP Hardening (LT-NGX-01), Thirdweb Engine EVM Address & BigInt Validation (LT-TWB-01).
- **v2.8.1 (Completed):** Cryptographic Sentinel Nonce Serialization (LT-POW-01), Strict SPV Merkle Proof Validation (LT-MRK-01), SECP256k1 Curve Point Validation & 33-Byte Compressed Parity (LT-CRY-01), Cross-Chain Canonical Address Ingress Normalization (LT-SOL-02), Multi-Core Parallel Thread Mining & Honest Telemetry (LT-GPU-01), and Master 100k Bot Army / Sybil Testnet Benchmark Suite.
- **v2.8.2 (Completed):** Zero-Bypass Public Key Address Derivation (rejection of off-curve/invalid keys with None without fallback spoofing), Strict Exception-Based Merkle Duplicate Rejection (CVE-2012-2459 with ValueError), Single-Transaction Block SPV Proof Parity (`proof_steps=[]` when `leaf == root`).
- **v2.8.3 (Completed):** Creator Follow Royalty On-Chain Indexing & UI Attribution, WrappedBCT Canonical Hash Double-Mint Immunity (`mapping(bytes32 => bool)`), PoW Single-Pass Execution & Worker Sync, Explorer Search DoS Defense (50-record cap), and Quick Compose Fee Synchronization (0.10 BCT).
- **v2.8.4 (Completed):** Systemd EnvironmentFile & Auto 256-bit Secret Provisioning (LT-SRV-01), URL Regex Isolation Sentinel (NEW-01), Coinbase Spot Price Feed CSP & Nginx Location Header Preservation (NEW-02 & NEW-03), Anti-Spoofing Quote Guard (NEW-04), WrappedBCT Un-Indexed Event Log Parity (NEW-05), Atomic Reference-Swap State Indexes (NEW-06), and Reply Regex Hyphenation & .env Inline Comment Parsing (NEW-07).
- **v2.8.5 (Completed):** Nginx Health Endpoint Proxy Rate Limiting & Real IP Forwarding (REV11-01), WrappedBCT Mapping Parity (REV11-02), Thirdweb Sensitive Secret Masking (REV11-03), SPV Proof Depth Bounds (REV11-04), WrappedBCT Ownable2Step Governance (REV11-05), Dynamic OpenCL Clamping & HTTP Timeouts (REV11-06).
- **v2.8.6 (Completed):** PoW Zero-Allocation Leading Byte Checks (AUD12-01), Thirdweb Secret Redaction (AUD12-02), WrappedBCT Gas SSTORE Optimization (AUD12-03), Web Push Length Bounds & Bounded Thread Pools (AUD12-04), Mnemonic BIP-39 NFKD Normalization (AUD12-05), Compact Block Invariant Verification (AUD12-06), Shared Miner Engine Singleton (AUD12-07), Nginx HTTP CSP Hardening (AUD12-08).
- **v2.8.7 (Completed):** Thirdweb CLI Process Exit Clean Termination (AUD14-01), SECP256k1 Curve Point Single-Pass Decompression (AUD14-02), PoW Nonce Offset & SHA-256 Midstate Stream Cloning (AUD14-03), Frontend DOM XSS Attribute Sanitization & Escaped Market Feeds (AUD14-04), Automated Production Rust maturin Build & Systemd LimitNOFILE (AUD14-05), Nginx Wallet Endpoint Request Body & Timeout Bounds (AUD14-06), SSH Log Inspector Timeout & Credential Masking (AUD14-07), Mnemonic Case Normalization (AUD13-01).
- **v2.8.8 (Completed):** Frontend Embedding Text Slicing Bug Fix (AUD15-01), Secret Masking Regex Callable & Dual DB Fallback (AUD15-02), Checked Arithmetic Integer Overflow Immunization (AUD15-03), Nginx Child Location CSP Inheritance & /integrity.json Parity (AUD15-04), Full BIP-39 NFKD Normalization Prior to Validation (AUD15-05), P2P Compact Block Version, Raw JSON & Short ID Verification (AUD15-06), Thirdweb 0x Key Normalization & CLI Flag Injection Defense (AUD15-07), Audit Bundle Privacy Filters & Dynamic Protocol SSOT (AUD15-08).
- **v2.8.9 (Completed):** High Start-Nonce Rayon Clamping (AUD16-01), Prioritized Non-Empty Database Discovery (AUD16-02), DOM XSS Attribute Sanitization (AUD16-03), P2P Compact Block Header & Duplicate ID Defense (AUD16-04), Thirdweb 64-Hex Validation & Dual-Prefix Key Redaction (AUD16-05), BIP-39 Pre-Split NFKD Unicode Normalization (AUD16-06), Audit Bundle Privacy Filters & Output XML Parity (AUD16-07), Universal Protocol SSOT Version Synchronization (AUD16-08).
- **v2.9.0 (Completed):** P2P getblocktxn Rate-Limit & Index Amplification Defense (AUD17-01), State Snapshot Memoization & Rate-Limit Guard (AUD17-02), P2P Gossip HTTP Redirect SSRF Protection (AUD17-03), Compact Block Miner Signature Forwarding (AUD17-04), Nginx /p2p/ Reverse-Proxy Routing (AUD17-05), Validator Stake/Unstake Rate-Limiting (AUD17-06), Authenticated Web Push Unsubscribe (AUD17-07).

- **v2.9.1 (Completed):** Bridge Gas Drain Prevention & Contract Deduplication Check (AUD17B-01), Rust PoW u64 Overflow-Free Ceiling Division (AUD17B-02), Wallet 64-Hex & SECP256k1 Curve Order Validation (AUD17B-03), Bridge Burn Queue Unrecoverable HTTP 400/422 Drain Protection (AUD17B-04), Bridge Relayer NoneType & FD Leak Guards (AUD17B-05), Prioritized Data DB Path & Decimal Loss Elimination (AUD17B-06), Netlify /p2p/* Forwarding Rule (AUD17B-07).
- **v2.9.2 (Completed):** Web3 v6 Compatibility & POA Middleware Cleanup (AUD18-01), Bridge Confirmation Deferral against Premature FAILED Locks (AUD18-02), Rust PoW Inclusive Range & u64::MAX Boundary Support (AUD18-03), Bridge Relayer 0x Canonical Prefix & Gas Defense (AUD18-04), BIP-39 NFKD Unicode Normalization & Order Consolidation (AUD18-05), Nginx Permissions-Policy Child Location Inheritance (AUD18-06), Audit Packaging & SSOT Protocol Synchronization (AUD18-07).
- **v2.9.3 (Current Production Release):** Bridge Queue Anti-Deadlock on Invalid/Unsealed Orders (AUD19-01), Bridge Burn Decimal Precision & 1 wBCT Minimum Anti-Dust Limit (AUD19-02), Nginx default_type Content-Type De-duplication (AUD19-03), Whitepaper PDF Generator Universal Header Filter (AUD19-04), Local Node Sync File List Expansion (AUD19-05), Double SHA-256 & Constructor Parameter Doc Parity (AUD19-06), Repomix Configuration & Base64 SVG Asset Filter (AUD19-07).
- **v3.0+ (Future Research):** Optional zk-SNARK shielded transfers, decentralized encrypted direct messaging (E2EE ECDH over SECP256k1), and native cross-chain HTLC atomic swaps.

References

Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System.*

Koblitz, N. (1987). *Elliptic Curve Cryptosystems.* Mathematics of Computation, 48(177), 203–209.

Pornin, T. (2013). *RFC 6979: Deterministic Usage of the Digital Signature Algorithm (DSA) and ECDSA.* IETF.

Palatinus, M., et al. (2013). *BIP-0039: Mnemonic Code for Generating Deterministic Keys.*

Corallo, M. (2016). *BIP-0152: Compact Block Relay.*

Castro, M., & Liskov, B. (2002). *Practical Byzantine Fault Tolerance and Proactive Recovery.* ACM TOCS.

Merkle, R. C. (1987). *A Digital Signature Based on a Conventional Encryption Function.* CRYPTO '87.